

TDDD25: Distributed Systems Programming Project

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Computer and Information Science
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Contact

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Organization

<https://www.ida.liu.se/~TDDD25/>

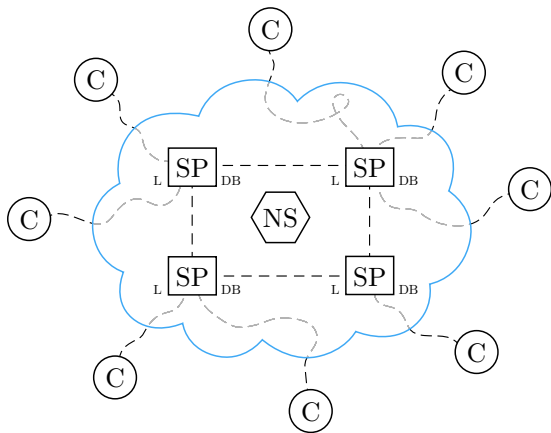
- 1 teaching session
- 7 lab sessions
- 1 + 5 labs
- 2 groups : Group *A* and Group *B*
- Registration deadline: January 24, 2021
- Completion deadline: March 10 – last lab session for your group

COVID-19 related info

<https://www.ida.liu.se/~TDDD25/>

- Lesson held in Zoom
 - link sent to all registered students of the course to their LiU e-mail
 - needs a Zoom account (LiU or otherwise) to access meeting
- Labs sessions held in Microsoft Teams
 - register to the lab in Webreg
 - all registered lab students are added to the same Microsoft Teams group(team) – global channel
 - each webreg subgroup of 2 students will have their own private Microsoft Teams subgroup for questions/demonstrations
 - write in global channel that you need assistance/want to demonstrate and write your private subgroup number
 - requests in the global channel will be handled in order of posting
- Remote work – check lab info
 - RDP client
 - Thinlinc
 - ssh

Distributed Database



- C — client
- SP — server/peer
- NS — name service
- DB — database
- L — lock

Source Code

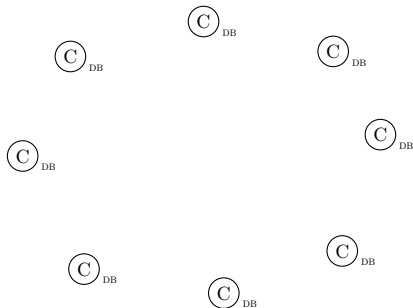
<https://gitlab.liu.se/LiU-TDDD25/labs/>

- doc/
- src/
 - lab0/
 - lab1/
 - lab2/
 - lab3/
 - lab4/
 - lab5/
 - modules/
 - Common/
 - Server/

Implementation

- Multi-threaded object-oriented code in Python 3
- Communication via objects serialized in JSON
- Data transfer through TCP sockets

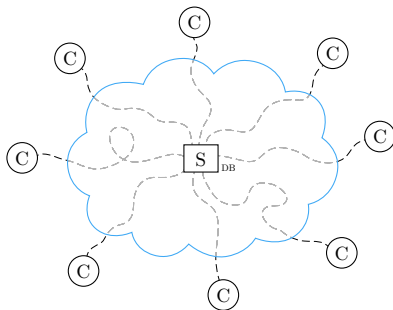
Lab 0: Standalone Database



- Local database for each client (C)
- **TODO**: complete the implementation of the read and write operations of the database (DB)

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab0.pdf>

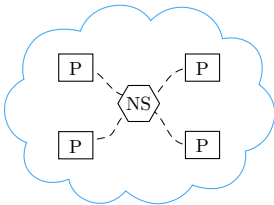
Lab 1: Client-Server Database



- Centralized database (DB)
- **TODO**: complete the implementation of the client (C) and server (S) communication mechanism

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab1.pdf>

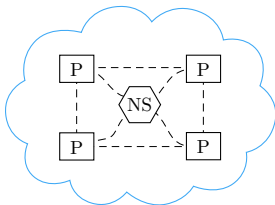
Lab 2: Object Request Broker



- Name service (NS) and object request broker (ORB)
- Abstract away the communication part of the functionality
- **TODO:** complete the implementation of the ORB

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab2.pdf>

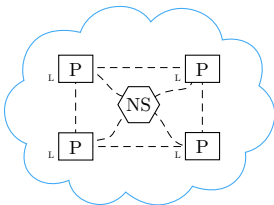
Lab 3: Peer-to-Peer Communication



- Smart mechanism for keeping track of peers (P)
- **TODO**: complete the functionality dealing with the peers joining or leaving the system

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab3.pdf>

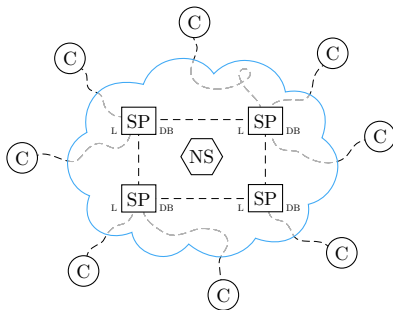
Lab 4: Distributed Locks



- Distributed mutual exclusion to control concurrent operations
- **TODO**: complete the implementation of the second Ricart–Agrawala algorithm (L)

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab4.pdf>

Lab 5: Client-Server Database with Replicas



- Everything together
- **TODO**: complete the implementation of the server/peer (SP) using all the previously developed components

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab5.pdf>

Submission

- No written reports are needed
- Make your code available on GitLab
- Add your teaching assistant as a Reporter
- Demonstrate your solutions in class
- **Attend your group's lab sessions**

Good luck!